

```
'wordpress' }
- { regexp: 'username_here', replace: 'guest' }
- { regexp: 'password_here', replace: 'osfy' }
- name: Restart apache2 server
service:
  name: apache2
  state: restarted
```

We create variables to store the downloaded file for WordPress, and the target installation path. After updating the software repository, a database is created for the WordPress application. The download and target directories are created on the guest system, before actually downloading the latest WordPress sources. A configuration file is then created, and the database settings are updated. Although we explicitly specify the password here, the recommended practice is to store the encrypted passwords in an Ansible Vault file, and reference the same in the playbook. In future articles, I will demonstrate this use case. After completing the configuration, the Web server is restarted. An execution run of the playbook is shown below:

```
$ ansible-playbook -i inventory/kvm/inventory playbooks/
configuration/wordpress.yml -k
SUDO password:
PLAY [Setup Wordpress] *****
TASK [setup] *****
ok: [ubuntu]
TASK [Update the software package repository] *****
changed: [ubuntu]
TASK [Create a database for wordpress] *****
changed: [ubuntu]
TASK [Create downloads directory] *****
ok: [ubuntu]
TASK [Create target directory] *****
changed: [ubuntu]
TASK [Download latest wordpress] *****
ok: [ubuntu]
TASK [Extract to /var/www/html] *****
changed: [ubuntu]
TASK [Copy wp-config-sample.php to wp-config.php]
changed: [ubuntu]
TASK [Update database credentials in the file] *****
changed: [ubuntu] => (item={u'regexp': u'database_name_here',
u'replace': u'wordpress'})
changed: [ubuntu] => (item={u'regexp': u'username_here',
u'replace': u'guest'})
changed: [ubuntu] => (item={u'regexp': u'password_here',
u'replace': u'osfy'})
```

```
TASK [Restart apache2 server] *****
changed: [ubuntu]
PLAY RECAP *****
ubuntu : ok=10 changed=7
unreachable=0 failed=0
```

If you open the URL <http://192.168.122.250/wordpress> in a browser on the host system, you will see a screenshot as shown in Figure 2.



Figure 2: WordPress install page

You can now proceed to complete the installation process from the browser. It is recommended that you follow the security best practices as recommended by the WordPress and PHP projects to harden this deployment.

Writing clean-up playbooks

It is essential to write clean-up playbooks to revert whatever changes you have made, so that you can roll back the system if things fail. Uninstalling should be done in the reverse order. For example, remove WordPress first, followed by PHP, MySQL and Apache.

The removal of WordPress could depend on your data retention policy. You might want to back up your PHP files, or you may decide to discard them. You might also want to retain the database. A complete removal of WordPress and the LAMP stack in the playbooks/admin folder is provided below for reference:

```
---
- name: Uninstall Wordpress
  hosts: ubuntu
```